Chapter 54. Psychological Problems in the Ventilated Patient

Yoanna Skrobik

Psychological Problems in the Ventilated Patient: Introduction

Psychological and psychiatric disturbances occur frequently in mechanically ventilated patients, and cause patients, their loved ones, and their caregivers considerable distress. This chapter familiarizes the reader with commonly observed psychological and psychiatric symptoms in ventilated critically ill patients; briefly discusses the challenge of making a psychiatric diagnosis in the context of the ventilated patient; and describes what is known of the three disorders most commonly described in this patient population: delirium, depression, and posttraumatic stress disorder.

Traditional psychological and psychiatric assessments in the adult patient rely on the patient's ability to engage in conversation. Baseline psychological health and personality, a patient's relationship with his or her acute or chronic illness, a patient's emotional response to it, and the clinical context in which the patient is placed are typically considered in this evaluation. Assessment of the psychological and psychiatric state of a mechanically ventilated patient is hindered by the presence of an endotracheal tube, which makes verbal communication impossible. A tracheotomy with a talking valve, and a face mask, also limits speech. Beyond problems of verbal communication, the sudden and dramatic onset of a ventilated patient's illness, the use of pharmacologic sedation, the layout of an intensive care unit (ICU), and a critical care physician's unfamiliarity with psychological assessments have hampered descriptions of psychiatric and psychological issues.

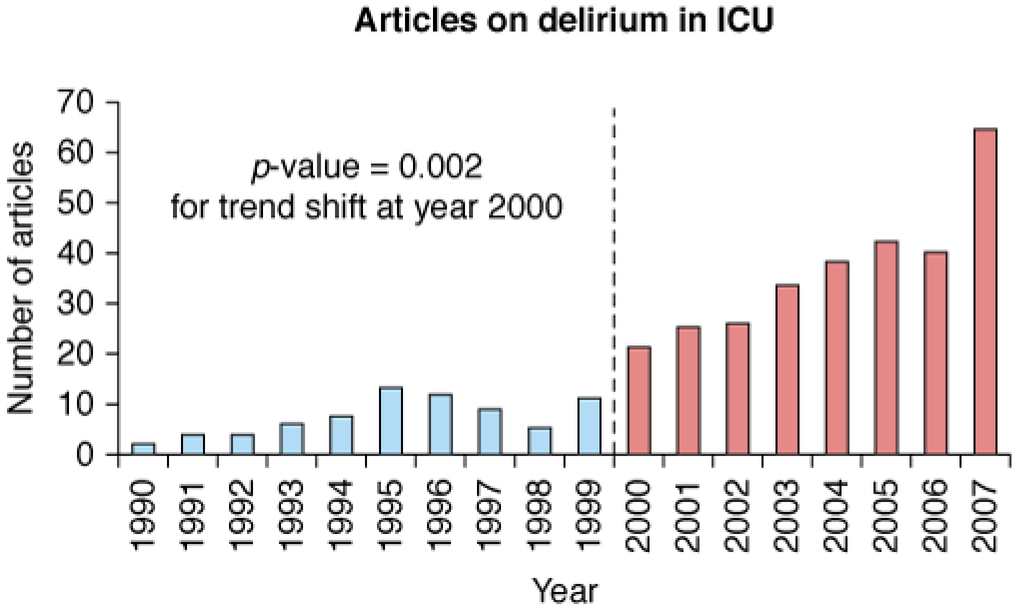
Psychiatric diagnoses are made on the basis of clinical criteria. These criteria were, for many decades, highly variable depending on a school of thought and geographical provenance. Lobotomies, incremental electroshock therapy to re-create neuropsychiatric normalcy, and insulin-coma treatments exposed psychiatric professionals to scientific and public criticism. Over the last 40 years, psychiatric diagnoses have been structured more rigorously by experienced psychiatrists based on symptom and disease classifications through the use of systems such as the *International Classification of Diseases* (ICD) from the World Health Organization, and the *Diagnostic and Statistical Manual of Mental Disorders* (DSM) from the American Psychiatric Association. The tenth revision of the *International Statistical Classification of Diseases and Related Health Problems* (ICD-10) was published in 1991; the fourth revision of the *Diagnostic and Statistical*

Manual of Mental Disorders, DSM-IV-TR*,* was published in 2000. The ICD-11 is expected in 2011 and the DSM-V update is anticipated in 2013.

The ICD-10 and the DSM-IV-TR rely on diagnostic criteria compiled by expert psychiatrists and, in the case of the DSM-IV*,* on the basis of described and published patient symptoms. Two problems become apparent when applying these diagnostic criteria to critically ill patients. The first is the challenge of applying diagnostic criteria to a mechanically ventilated patient if the symptom constellations have been gathered in a different—usually ambulatory—population. Apathy, for instance, a feature of depression, may be expected in a septic ventilated patient receiving propofol, making its value as a diagnostic criterion of depression in this context uncertain. The second is that these DSM or ICD criteria have not been investigated in terms of validity in ventilated patients, whether acutely or chronically ill, nor are the criteria correlated with outcomes in such patients.

Psychiatric or psychological disturbances in the ventilated critically ill patients have been given many names, such as ICU psychosis, septic encephalopathy, and apathy. Semantic descriptions for disorders such as delirium[1] vary by geographic region. Over the last decade, publications on psychiatric topics such as delirium in ventilated patients have increased significantly (Fig. 54-1). Two factors may have influenced this change. Two intensivist-led publications in 2001[2,3] sensitized clinicians to the notion of screening ventilated patients for delirium through the use of simple screening tool, applicable at the bedside. More delirium-related publications followed, with subsequent interest in other psychiatric and psychological disorders, such as depression and posttraumatic stress disorder. Almost simultaneously, researchers with an interest in ICU outcomes in adult respiratory distress syndrome (ARDS), whose focus was initially on respiratory physiology, realized the main impediments to recovery for these patients were myopathy and psychological distress.[4,5] Three psychiatric syndromes—delirium, depression, and posttraumatic stress disorder—have since been identified with increasing frequency [6] in ventilated critically ill patients and ICU survivors.[7] The associations between these diagnoses and poor outcomes, including length of stay, quality of life, and cognitive impairment, have led to a cry for pharmacologic intervention for delirium by interventionists, and for further longitudinal studies[8,9] by those interested in the sequelae of what had traditionally believed to be rapidly reversible (and almost irrelevant) disorders.

Figure 54-1



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

The number of publications on delirium in the critical care setting (in mechanically ventilated and nonmechanically ventilated patients) over 17 years. (Modified, with permission, from Morandi A, Pandharipande P, Trabucchi M, et al. Understanding international differences in terminology for delirium and other types of acute brain dysfunction in critically ill patients. *Intensive Care Med.* 2008;34:1907–1915.)

What is understood of these three psychiatric syndromes is limited by the dearth and relative novelty of literature on the topic.[10] Current knowledge should be considered in light of the methodological limitations in establishing a diagnosis as described above.

Most publications reviewed for this chapter are focused exclusively or largely on mechanically ventilated patients, a group that some authors argue is particularly susceptible to psychiatric complications, such as delirium.[11] Information from nonventilated patients is discussed if it is consistent with what has been published on ventilated patients, but only if it provides insight not available from the literature on mechanically ventilated patients per se. Reviewed studies are limited to critical care and stepdown units; studies in patients receiving home ventilation or chronic ventilation on a hospital ward or a specialized institution were not considered. The disproportion between the delirium literature in critical care and that describing other syndromes, which is more modest, explains the discrepancy in content herein.

Diagnostic and Contextual Challenges

Unsurprisingly, publications on the subject of psychological and psychiatric disorders in ICU patients have been driven by the medical perspective of the psychiatric diagnosis. Most physicians attempt to classify psychological or psychiatric symptoms as diseases, with attributable biologic alterations and reproducible symptoms. Medical culture[12] drives a rigid medical approach, and a rationalization of “disease”; the narrow view of biologic alterations and their diagnostic criteria is the focus of what we publish and read. Dimensions such as illness (the patients' perception of the self) and sickness (the social dimension and impact of the diagnosis, particularly a psychiatric one) are seldom addressed. In settings outside critical care, caregivers understand that a patient's well-being depends on psychological integrity. Such well-being, and perhaps even the will to survive, are hindered if the patient is unable to recognize his or her own self during disease, or if that very self is compromised, a common occurrence in the foreign, disturbing, and overwhelmingly technological setting of the ICU. This wretchedness is complicated by a complete dependence on machines and personnel. What little autonomy a patient retains is seldom fostered or encouraged. The relationship with the caregiver, or a significant person in the patient's life, also contributes to healing, or to a more serene journey toward death. The social impact of a psychological or psychiatric disease may interfere with this patient–caregiver or patient–significant person relationship, and constitute a dimension of "sickness." Establishing a diagnosis of depression and delirium requires descriptors; the dimensions of illness and sickness, however, may be what cause the greatest distress. Patient recovery and postdischarge outcomes may be related to these dimensions of illness and sickness; however, our limited understanding of these facets as experienced by patients, their caregivers, and loved ones limits a global understanding of the psychological and psychiatric issues and precludes a holistic approach to care.

Furthermore, context is important in establishing whether endogenous or exogenous factors trigger psychiatric or psychological disturbances. Some psychiatric diagnoses can be made with clarity. For instance, mania in an ambulatory outpatient has in many cases a recognizable pattern, associated biochemical abnormalities, and a predictable response to medication. Syndromes such as depression can come from within, or be considered a normal response (in grieving, for example). Other mood states such as anxiety range from normal to pathologic, and have not been defined in the critically ill. Several symptoms (sadness, paranoid delusions, or harrowing flashbacks) could, arguably, be considered contingent, temporary, and normal responses to painful, traumatic, or near-death experiences. Overlapping syndromes of delirium and depression have been described.[13,14] The administration of psychotropic drugs in high doses over prolonged periods of time adds an important confounder, as do personality traits and the amount of cognitive reserve. This issue of context makes the attribution of a psychiatric diagnosis to a set of symptoms particularly challenging in the critical care setting. Dialog is a key element in differentiating these factors as potential contributors, in establishing the diagnoses, and in proffering expression as a form of early therapeutic intervention. Verbal dialog is challenging in the context of mechanical ventilation. Weakness, which precludes movement, facial expression, or both, further challenges (or is perceived as challenging) the examiner and contributes to the real or perceived barriers to communication.

The three most commonly described psychiatric syndromes in ICU patients and survivors are delirium, depression, and posttraumatic stress disorder. Each is discussed in turn.

Delirium

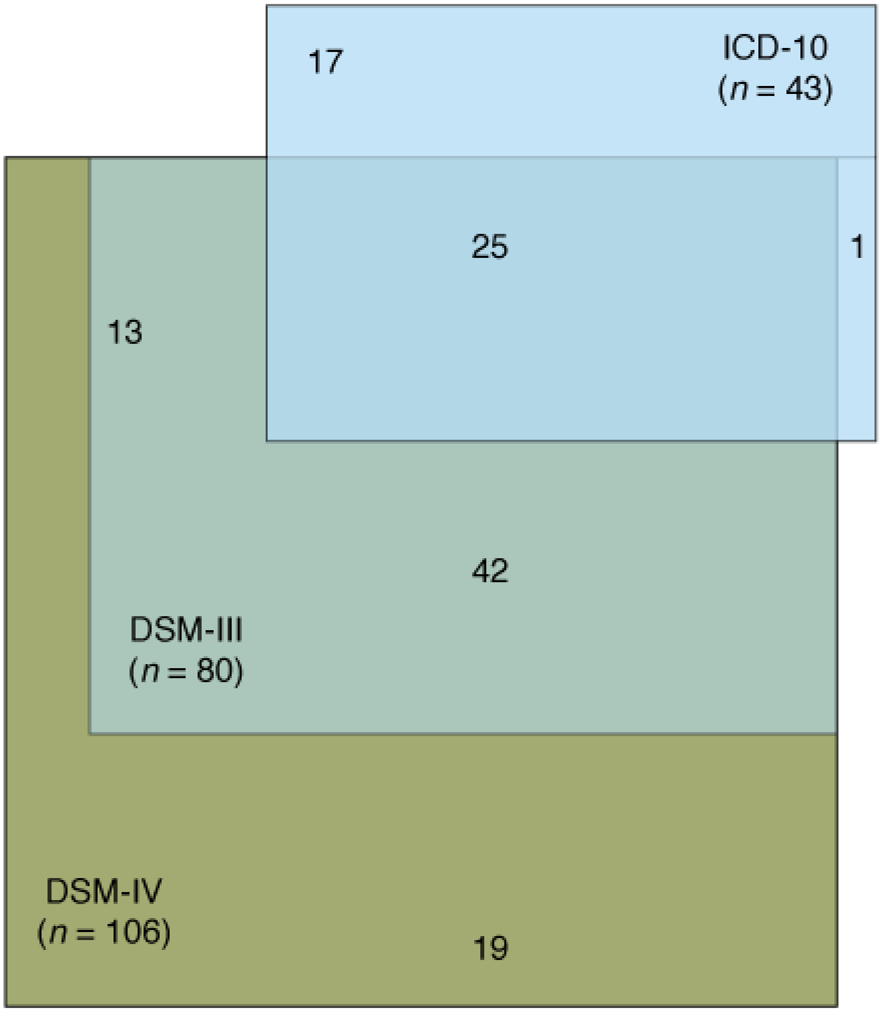
The patient you spoke to and communicated with yesterday on ICU rounds is a little more difficult to rouse this morning. When you approach him, he initially looks surprised and frightened; as you introduce yourself and remind him of yesterday's conversation, he appears uncertain of who you are and of his surroundings. As you continue speaking, he looks away and appears to stare at something on the ceiling. The nurse reports that a few hours earlier he was grasping at what appeared to be imaginary objects in front of him. He did not sleep last night.

Delirium, with its traditional symptoms described here, frightens patients. The delusions and hallucinations they experience transiently and usually for the first time in their lives are upsetting; paranoid thoughts are common. When cognitive function normalizes in the fluctuating course that characterizes delirium, patients wonder if they are losing their minds and if they will ever return to the way they were before. Because 40% of patients present symptoms only between midnight and 8 am, careful attention to the nurse's report of a patient's behavior is helpful in recognizing the problem.

Diagnosis

A diagnosis of delirium is based on the reference standard of symptoms described in nonventilated patients deemed clinically delirious, mostly among geriatric patients where the syndrome is prevalent. The DSM-IV criteria require a fluctuating disturbance in cognition and in consciousness in the context of an acute medical illness. Conversely the World Health Organization's ICD-10[15] includes different criteria for clinical and research use, both of which include a broad range of features that capture the phenomenologic complexity of the syndrome. Comparison of these two sets of criteria in a group of 425 nonventilated elderly medical or nursing home patients showed that applying the DSM-IV criteria identified more (n = 106) patients as having delirium than did the ICD-10 criteria, with twenty-five patients considered delirious by both sets of diagnostic criteria (Fig. 54-2).[16] Surprisingly, this study reported similar outcomes, in terms of mortality or length of stay, regardless of the criteria applied to diagnose delirium (Table 54-1).

Figure 54-2



Source: Tobin MJ: *Principles and Practice of Mechanical Ventilation, 3rd Edition:* www.accessanesthesiology.com

Comparison of different screening tools in the same population of 425 nonmechanically ventilated elderly medical or nursing home patients using DSM-IV criteria and earlier DSM criteria (DSM-III) as well as ICD-10 criteria. More (n = 106) patients were identified as having delirium with the DSM-IV than with any other set of diagnostic criteria. (Adapted, with permission, from Laurila JV, Pitkala KH, Strandberg TE, Tilvis RS. Impact of different diagnostic criteria on prognosis of delirium: a prospective study. *Dement Geriatr Cogn Disord.* 2004;18:240–244.)

Table 54-1: Outcomes (Mortality or Length Of Stay) in Relationship to the Criteria (DSM IV, DSM III, and ICD-10) Used to Diagnose Delirium in 425 Nonmechanically Ventilated Elderly Medical or Nursing Home Patients (Outcomes Were Similar)

Prognostic Variable	DSM-IV (*n* = 106)	DSM-III (*n* = 80)	ICD-10 (*n* = 43)	*p*-value[a]
Mortality/1 year, %	34.9	36.3	41.9	n.s.
Mortality/2 year, %	58.5	62.5	65.1	n.s.
Mean days in institutions/1 year[b]	220(150)	230(148)	211(157)	n.s.
New admissions to permanent institutional care/2 year, %	55.0[33/60]	60.0[24/40]	64.3[9/14]	n.s.
In permanent institutional care or deceased/2 year, %	90.6	93.8	97.7	n.s.

n.s. = Not significant. Figures in parentheses indicate standard deviations, those in brackets actual number institutionalized/total number from community care, *y* care.

[a]Proportions and their differences were tested by the χ^2 test, d.f. = 3.

[b]Means and their differences were tested by analysis of variance.

Source: Adapted, with permission, from Laurila JV, Pitkala KH, Strandberg TE, Tilvis RS. Impact of different diagnostic criteria on prognosis of delirium: a prospective study. *Dement Geriatr Cogn Disord.* 2004;18:240–244.

Thorough descriptions of the natural history of delirium in ventilated patients are lacking, as is a good understanding of its basic scientific and biologic mechanisms. In 2001, two groups reported independently validated delirium screening scales for use in ventilated critically ill patients.[3,4] The first, the Intensive Care Delirium Screening Checklist (ICDSC), is an eight-point item scale, where four points or more correspond to a clinical diagnosis of delirium. The scale was validated against the clinical bedside opinion of a psychiatrist. The advantage of using this eight-item scale with a four-point cutoff is its ability to detect subsyndromal delirium in patients with scores of more than 0 but less than 4 items. Subsyndromal delirium is considered clinically significant by psychiatrists[17,18] and, in ventilated and nonventilated ICU patients, predicts an intermediate risk of prolonged length of stay and mortality when compared with asymptomatic patients,[19] who did better, or delirious patients, whose prognosis was the worst. The second delirium scale was originally adapted as a simplification of the DSM-IV–based Confusion Assessment Method (CAM) scale, so as to make it applicable to ventilated patients; it was validated against the clinical opinion of a geriatrician. This modified scale, the Confusion Assessment Method in Intensive Care Units (CAM-ICU),[2] is binary. Other scales, such as the Delirium Detection Scale (DDS) and the Nursing Delirium Screening Scale (NuDESC), have also been assessed in critically ill patients; only the ICDSC and the CAM-ICU are presented here because of their broad application to and validation in ventilated patients, and because of their psychometric quality.

Clinicians are reported to underrecognize delirium.[20,21] This has led some critical care professional societies to promote routine critical care delirium screening.[22] Canadian governing bodies mandate it for hospital accreditation. Perusal of the literature on screening tools, however, shows that these validated screening tools yield a broad distribution of incidences of this condition. Studies describing the binary CAM-ICU in similar populations reveal delirium incidences that range from 10%[23] to over 80%. Sedation may confound the scale measurements.[24] In contrast, the range in reported delirium rates using the ICDSC is somewhat narrower: 32%[25] to 45%.[21] In addition to grading cognitive normalcy, subsyndromal delirium and delirium the ICDSC describes specific symptoms,[26] and with them, the prognosis conferred by each one. Whether these two tools screen for the same constellation of symptoms is unclear, given the conflicting results in the publications on the subject.[27,28] Because of the numerous methodological issues described above, comparisons of sensitivity and specificity of delirium screening scales to mechanically ventilated patients in and outside the ICU are difficult to address.

Awareness as to which patient is more likely to develop delirium in the ICU will identify patients most likely to benefit from preventive strategies and alert caregivers as to which patients will require more thorough evaluation for delirium symptoms. These risk factors for delirium cannot identify patients who will respond to therapeutic intervention. Different clinical features have been associated with a higher incidence of delirium.[29] The incremental likelihood of delirium symptoms increases with severity of illness and with excessive alcohol consumption.[29] Hypertensive patients are more likely to become delirious in the ICU,[25] as are patients with preexisting dementia and patients who are heavily sedated.[25,29] In contrast to patients admitted to a hospital ward, age is not consistently related to the likelihood of developing delirium among ventilated patients,[25,30] although there may be an association with age in mostly nonventilated patients older than 65 years when the CAM-ICU delirium detection tool is used.[31,32]

There are significant gaps in our knowledge regarding delirium in ventilated patients. Despite reports that mechanical ventilation confers a risk for delirium,[33] the question of whether the association is with mechanical ventilation per se or an independent feature linked to severity of illness or drugs administered has not been addressed in studies where known risk factors are taken into account. Cardiovascular patients, who are burdened with cognitive dysfunction after cardiac surgery and ICU discharge, remain understudied, as do neurologic patients, and trauma patients (including those with traumatic brain injury), perhaps because of the inherent difficulties in making an assessment as a result of confounding clinical features.

Prevention

Early physiotherapy and mobilization, when implemented to aid myopathy, significantly reduce delirium rates.[34,35] These results raise the question of whether nonpharmacologic interventions, which encourage patients to focus on their autonomy, may prevent or alleviate delirium. Sedatives and analgesics, when carefully titrated and administered according to symptoms, are associated with lower rates of subsyndromal delirium and an increase in the probability of a patient being able to return home.[36] Whether the particular type of sedative agent makes any difference to the probability of delirium is not clear. Once delirium develops, however, continuous sedation with dexmedetomidine is associated with lower duration of delirium than continuous intravenous sedation with midazolam, a short-acting benzodiazepine,[37] in medical and surgical patients. It is not clear whether these results reflect an inherent problem with midazolam infusions or a therapeutic benefit of dexmedetomidine.[38] In cardiac surgery patients, the use of propofol as a sedative did not alter the risk of delirium as compared with dexmedetomidine.[39]

Treatment

Although administration of antipsychotics has been a mainstay in the pharmacologic management of delirium in critically ill patients,[40] there is no scientific basis for this practice or evidence of benefit of administering antipsychotic agents in delirious critically ill ventilated patients[41]: neither the duration of delirium nor its severity is reduced. A possible exception is the atypical antipsychotic, quetiapine,[42] which, in contrast to other pharmacologic agents,[8,41] produced a reduction in the duration of ICU delirium, albeit in a single pilot study.

Descriptions by observers, and stories[43] told by individual patients,[44] describe the anguish, fear, and harrowing nature of patient perceptions in a delirious state. Some narratives and recall studies describe the positive impact of reassuring or reality-orienting caregivers[45] within the critical care setting. The added contribution made by visiting family members and loved ones[29] are in keeping with the positive impact of nurse-facilitated family participation in the care of a delirious patient.[46]

Some investigators have made links between delirium in the ICU and long-term cognitive dysfunction.[47] These data should be considered with caution given the frequency of cognitive dysfunction in nondelirious patients,[48] given our difficulties in establishing the diagnosis of delirium, and given its many potential confounders, such as the risk of dementia in patients older than 65 years, which doubles every 5 years between the ages of 65 and 85 years.

Depression

On morning rounds, the patient whose ICU stay was marred by multiple septic complications of his intraabdominal surgery does not look interested in your informing him he is finally ready to be extubated. After the extubation and despite no longer receiving sedatives, he is lying in bed immobile, and answers questions monosyllabically. He tells the nurse he is not interested in sitting up in a chair or in spirometry with the physiotherapist. No amount of encouragement seems to make a difference. His wife tells you he is “not himself,” and is concerned.

Is this patient depressed? Or is he reacting normally to a near-death experience during which his mood was affected by circulating inflammatory mediators and the sedatives he received? The patient’s hypoactive reaction and mood alteration is not unusual, especially because he was septic. A diagnosis of depression cannot be excluded, and the patient’s evolution should be monitored with careful attention to psychological symptoms. Depression is not reported in ventilated ICU patients; descriptions exist primarily in ICU survivors[49] and in patients who have been transferred from a critical care setting to a specialized center that focuses on weaning from prolonged mechanical ventilation.[50] To assess whether patients are depressed or at risk for depression, questionnaires, such as the Hospital Anxiety and Depression Scale (HADS),[51] have been used as substitutes or complements to clinical assessments.[52] Studies that describe clinical assessments, such as one-on-one interviews by an experienced caregiver (a psychologist or a psychiatrist) of ICU patients, weaning patients,[50] or survivors are rare. Depression in critically ill patients, whether mechanically ventilated or not, is associated with both the severity of illness on admission[53] and with the administration of benzodiazepines.[54] Observational studies are few, and potential confounders in making a diagnosis of depression are numerous.

Sickness behavior is a term coined to describe the behavioral changes that develop in ill individuals during the course of an infection. This sickness behavior,[55] described in sepsis, is associated with lethargy, sleepiness, inability to concentrate, and apathy. Whether these symptoms are a manifestation of underlying medical illness or related to depression per se is not clear.

Risk factors for developing depression in association with mechanical ventilation or ICU admission have not been well established; patients with different profiles may be at different risk and have a different prognosis. For instance, 28% of more than 66,000 veterans admitted to nonsurgical ICUs[56] had a history of an earlier psychiatric diagnosis, a feature associated with a greater likelihood of being subsequently diagnosed with depression or with posttraumatic stress disorder. Patients with diabetes who have experienced depression are more likely to end up in an ICU than are patients with diabetes who have no baseline psychiatric history.[57] ICU admission in patients with diabetes is independently associated with a higher probability of subsequent major depression.[58] These large descriptive studies included both ventilated and nonventilated patients. No validated assessment tools exist to screen for the presence or severity of depression in mechanically ventilated patients. In specialized critical care settings where psychologists are part of the ICU team, clinical assessments of depression are probably reliable. Most intensive care units do not include such experts. Longitudinal studies suggest post-ICU depression is frequent. Early expert screening may benefit patients and their familiesy. No clear epidemiologic descriptions of depression and its outcomes in ventilated patients have been published. Whether any particular form of therapy or none is preferable in the management of depression in ventilated patients is not clear from the literature to date.

Posttraumatic Stress Disorder

The patient you discharged from the ICU 5 weeks ago with epiglottitis comes to your clinic for a follow-up appointment. Since her return home, she describes waking up several times each night in a cold sweat from a recurrent dream that involves attempts at intubating her while she is developing an obstructed airway. She is jumpy and frightened when she goes about the simplest of her daily tasks, and lashed out at a grocery store attendant who asked her at the checkout whether she had change. Her sister confirms how unusual this behavior is for her, and expresses concern because the patient is not functional.

Posttraumatic stress disorder (PTSD) symptoms, such as those described here, is an anxiety disorder associated with a traumatic event that involved the threat of injury or death. Diagnostic symptoms for PTSD include reexperiencing the original trauma(s) through flashbacks or nightmares, avoidance of stimuli associated with the trauma, and increased arousal, such as difficulty in falling asleep or staying asleep, anger, and hypervigilance. Formal diagnostic criteria (both DSM-IV-TR and ICD-10) require that the symptoms persist for more than 1 month and cause significant impairment in social, occupational, or other important areas of functioning. Stressors believed to trigger PTSD include myocardial infarction and critical illness.

Increasing numbers of publications have addressed PTSD in mechanically ventilated and nonventilated ICU survivors[59,60] over the last decade. Its reported incidence varies from 8% to 51%. The amount of literature is low, particularly when one considers the frequency with which the disorder is diagnosed.

Questionnaire-based screening, telephone-based screening, and clinical assessments with and without questionnaires to aid in establishing the diagnosis have populated the studies to date. Questionnaires and clinical assessments, however, may not have the same correlation in mechanically ventilated critically ill patients as they do in other patients. When patients admitted to a weaning center (who had been or still were being mechanically ventilated) were assessed with a standardized PTSD questionnaire and a clinical assessment by a clinical psychologist,[61] the results revealed that the threshold on the questionnaire, which had been validated to predict PTSD in non-ICU patients, was in fact lower than the scoring threshold that corresponded to the clinical diagnosis of PTSD in ICU survivors. These data raise concerns about underestimating the prevalence of PTSD in ventilated patients if standard thresholds are applied.

The only study addressing the link between psychopathology in the ICU and PTSD did not document a link between delirium and PTSD, albeit in a cohort where the incidence of PTSD was low.[62]

PTSD seems to be a risk factor for fatal and nonfatal cardiac events after discharge from hospital.[33] ICU-related associations include lower factual recall and its corollary, higher sedative administration,[63] during the ICU stay.

One study suggests that early and targeted psychological intervention, consisting of five or six face-to-face meetings involving direct ICU patient counseling, stress management, support, and education by a clinical psychologist,[64] offers better outcomes from PTSD than standard ICU care.

What remains surprising is that this condition, which requires a profound precipitant, commonly of a life-threatening magnitude, has been so poorly studied in critically ill patients where exposure to this type of stressor is so predictable.

Nonpharmacologic Interventions

Nonmedical interventions can be helpful in many psychiatric and psychological disturbances. Outside the ICU, prevention of syndromes such as delirium through the use of interventions such as rehydration, reorientation, and warm milk and massages in the evening can sometimes yield spectacular results[65] and are extremely cost-effective. Physiotherapy and early mobilization certainly appear to have a beneficial impact on ICU delirium.[34,35] Interventions such as music may help, and are extremely unlikely to do harm.[66] A bedside diary may aid the most severely traumatized survivor make sense of his or her experience.[67] Finally, opportunities for ICU survivors to tell their tale in a structured fashion may be therapeutic.[68] The narrative has also been described as a tool for healing in the context of illness[69,70] and as a crucial instrument in transforming suffering for the individual who is telling the tale.[71]

Which type of follow-up care, with narratives or without, would best profit ICU survivors is uncertain,[72] although routine screening after ICU discharge and as-needed referral services may be of benefit.[73,74]

Summary

Psychological and psychiatric disturbances are extremely common in the mechanically ventilated critically ill patient. The natural history of such disturbances is not well understood. To date, three disorders have been described in some detail: delirium, depression, and PTSD. Nonpharmacologic approaches, such as early mobilization, adjustment and minimization of sedation, and psychological support, appear to offer some promise. No pharmacologic approach can be endorsed unequivocally.

Conclusion

Mechanically ventilated patients are now the subject of more focused investigations with regard to their psychological and psychiatric well-being, both in the acute setting and in terms of outcomes. Regardless of the specific psychological disturbance or psychiatric diagnosis, there appears to be a link between psychological abnormalities and the severity of illness.[75] The probability of developing psychological or psychiatric complications may depend on vulnerabilities such as prior psychiatric diagnosis[76] or variability in what is referred to as cognitive reserve. Although the severity of illness on ICU admission and some clinical features, such as excessive alcohol consumption, are associated with the development of delirium, the risks or associations with other disorders are not known. Diagnostic categories (delirium, depression, and PTSD) have not been simultaneously and comprehensively compared with patient outcomes. Less discrete but important syndromes, such as adjustment disorder, generalized anxiety, and panic disorder, remained unexplored in this population. Better, more thorough, and more holistic explanations and approaches await future investigations.

References

1. Morandi A, Pandharipande P, Trabucchi M, et al. Understanding international differences in terminology for delirium and other types of acute brain dysfunction in critically ill patients. *Intensive Care Med.* 2008;34:1907–1915. [PubMed: 18563387]

2. Bergeron N, Dubois MJ, Dumont M, et al. Intensive care delirium screening checklist: evaluation of a new screening tool. *Intensive Care Med.* 2001;27:859–864. [PubMed: 11430542]

3. Ely EW, Inouye SK, Bernard GR, et al. Delirium in mechanically ventilated patients: validity and reliability of the confusion assessment method for the intensive care unit (CAM-ICU). *JAMA.* 2001;286:2703–2710. [PubMed: 11730446]

4. Herridge MS, Cheung AM, Tansey CM, et al. One-year outcomes in survivors of the acute respiratory distress syndrome. *N Engl J Med.* 2003;348:683–693. [PubMed: 12594312]

5. Herridge MS, Tansey CM, Matté A, et al. Functional disability 5 years after acute respiratory distress syndrome. *N Engl J Med.* 2011;364:1293–1304. [PubMed: 21470008]

6. Rattray JE, Hull AM. Emotional outcome after intensive care: literature review. *J Adv Nurs.* 2008;64:2–13. [PubMed: 18721158]

7. Scragg P, Jones A, Fauvel N. Psychological problems following ICU treatment. *Anaesthesia.* 2001;56:9–14. [PubMed: 11167429]

8. Girard TD, Pandharipande PP, Carson SS, et al. Feasibility, efficacy, and safety of antipsychotics for intensive care unit delirium: the MIND randomized, placebo-controlled trial. *Crit Care Med.* 2010;38:428–437. [PubMed: 20095068]

9. Jackson JC, Gordon SM, Hart RP, et al. The association between delirium and cognitive decline: a review of the empirical literature. *Neuropsychol Rev.* 2004;14:87–98. [PubMed: 15264710]

10. Weinert C. Epidemiology and treatment of psychiatric conditions that develop after critical illness. *Curr Opin Crit Care.* 2005;11:376–380. [PubMed: 16015119]

11. Ely EW, Shintani A, Truman B, et al. Delirium as a predictor of mortality in mechanically ventilated patients in the intensive care unit. *JAMA.* 2004;291:1753–1762. [PubMed: 15082703]

12. Good BJ. *How Medicine Constructs Its Objects. Medicine, Rationality, and Experience: an Anthropological Perspective.* New York, NY: Cambridge University Press; 1994.

13. Leonard M, Spiller J, Keen J, et al. Symptoms of depression and delirium assessed serially in palliative-care inpatients. *Psychosomatics.* 2009;50:506–514. [PubMed: 19855037]

14. Givens JL, Jones RN, Inouye SK. The overlap syndrome of depression and delirium in older hospitalized patients. *J Am Geriatr Soc.* 2009;57:1347–1353. [PubMed: 19558475]

15. World Health Organization, Mental and behavioral disorders (F00–F99). *The International Classification of Diseases,* 10th rev. ICD-10. Geneva, Switzerland: World Health Organization; 1992.

16. Laurila JV, Pitkala KH, Strandberg TE, Tilvis RS. Impact of different diagnostic criteria on prognosis of delirium: a prospective study. *Dement Geriatr Cogn Disord.* 2004;18:240–244. [PubMed: 15286453]

17. Cole M, McCusker J, Dendukuri N, Han L. The prognostic significance of subsyndromal delirium in elderly medical inpatients. *J Am Geriatr Soc.* 2003;51:754–760. [PubMed: 12757560]

18. Marcantonio ER, Kiely DK, Simon SE, et al. Outcomes of older people admitted to postacute facilities with delirium. *J Am Geriatr Soc.* 2005;53:963–969. [PubMed: 15935018]

19. Ouimet S, Riker R, Bergeron N, et al. Subsyndromal delirium in the ICU: evidence for a disease spectrum. *Intensive Care Med.* 2007; 33:1007–1013. [PubMed: 17404704]

20. Spronk PE, Riekerk B, Hofhuis J, Rommes JH. Occurrence of delirium is severely underestimated in the ICU during daily care. *Intensive Care Med.* 2009;35:1276–1280. [PubMed: 19350214]

21. Roberts B, Rickard CM, Rajbhandari D, et al. Multicentre study of delirium in ICU patients using a simple screening tool. *Aust Crit Care.* 2005;18:6, 8–9, 11-4 passim. [PubMed: 18038529]

22. Martin J, Heymann A, Bäsell K, et al. Evidence and consensus-based German guidelines for the management of analgesia, sedation and delirium in intensive care—short version. *Ger Med Sci.* 2010;8:Doc02.

23. van Eijk MM, Roes KC, Honing ML, et al. Effect of rivastigmine as an adjunct to usual care with haloperidol on duration of delirium and mortality in critically ill patients: a multicentre, double-blind, placebo-controlled randomised trial. *Lancet.* 2010;376:1829–1837.

24. Kress JP. The complex interplay between delirium, sepsis and sedation. *Crit Care.* 2010;14:164. [PubMed: 20587084]

25. Ouimet S, Kavanagh BP, Gottfried SB, Skrobik Y. Incidence, risk factors and consequences of ICU delirium. *Intensive Care Med.* 2007;33:66–73. [PubMed: 17102966]

26. Marquis F, Ouimet S, Riker R, et al. Individual delirium symptoms: do they matter? *Crit Care Med.* 2007;35:2533–2537. [PubMed: 18084841]

27. Riker RR, Robbins T, Bruce H, et al. ICU delirium assessment tools often disagree. *Crit Care Med.* 2006;34(12): A7.

28. Plaschke K, von Haken R, Scholz M, et al. Comparison of the confusion assessment method for the intensive care unit (CAM-ICU) with the Intensive Care Delirium Screening Checklist (ICDSC) for delirium in critical care patients gives high agreement rate(s). *Intensive Care Med.* 2008;34:431–436. [PubMed: 17994221]

29. Van Rompaey B, Elseviers MM, Schuurmans MJ, et al. Risk factors for delirium in intensive care patients: a prospective cohort study. *Crit Care.* 2009;13:R77.

30. de Rooij SE, Govers AC, Korevaar JC, et al. Cognitive, functional, and quality-of-life outcomes of patients aged 80 and older who survived at least 1 year after planned or unplanned surgery or medical intensive care treatment. *J Am Geriatr Soc.* 2008;56:816–822.

31. Pisani MA, Kong SY, Kasl SV, et al. Days of delirium are associated with 1-year mortality in an older intensive care unit population. *Am J Respir Crit Care Med.* 2009;180:1092–1097. [PubMed: 19745202]

32. Pisani MA, Murphy TE, Araujo KL, Van Ness PH. Factors associated with persistent delirium after intensive care unit admission in an older medical patient population. *J Crit Care.* 2010;25:540.e1–e7. [PubMed: 20413252]

33. Tsuruta R, Nakahara T, Miyauchi T, et al. Prevalence and associated factors for delirium in critically ill patients at a Japanese intensive care unit. *Gen Hosp Psychiatry.* 2010;32:607–611. [PubMed: 21112452]

34. Schweickert WD, Pohlman MC, Pohlman AS, et al. Early physical and occupational therapy in mechanically ventilated, critically ill patients: a randomised controlled trial. *Lancet.* 2009;373:1874–1882. [PubMed: 19446324]

35. Needham DM, Korupolu R, Zanni JM, et al. Early physical medicine and rehabilitation for patients with acute respiratory failure: a quality improvement project. *Arch Phys Med Rehabil.* 2010;91:536–542. [PubMed: 20382284]

36. Skrobik Y, Ahern S, Leblanc M, et al. Protocolized intensive care unit management of analgesia, sedation, and delirium improves analgesia and subsyndromal delirium rates. *Anesth Analg.* 2010;111:451–463.

[PubMed: 20375300]

37. Riker RR, Shehabi Y, Bokesch PM, et al. Dexmedetomidine vs midazolam for sedation of critically ill patients: a randomized trial. *JAMA.* 2009;301:489–499. [PubMed: 19188334]

38. Maldonado JR, Wysong A, van der Starre PJ, et al. Dexmedetomidine and the reduction of postoperative delirium after cardiac surgery. *Psychosomatics.* 2009;50:206–217. [PubMed: 19567759]

39. Shehabi Y, Grant P, Wolfenden H, et al. Prevalence of delirium with dexmedetomidine compared with morphine based therapy after cardiac surgery: a randomized controlled trial (DEXmedetomidine COmpared to Morphine—DEXCOM Study). *Anesthesiology.* 2009; 111:1075–1084. [PubMed: 19786862]

40. Jacobi J, Fraser GL, Coursin DB, et al. Clinical practice guidelines for the sustained use of sedatives and analgesics in the critically ill adult. *Crit Care Med.* 2002;30:119–141. [PubMed: 11902253]

41. Devlin JW, Skrobik Y. Antipsychotics for the prevention and treatment of delirium in the intensive care unit: what is their role? *Harv Rev Psychiatry.* 2011;19:59–67. [PubMed: 21425934]

42. Devlin JW, Roberts RJ, Fong JJ, et al. Efficacy and safety of quetiapine in critically ill patients with delirium: a prospective, multicenter, randomized, double-blind, placebo-controlled pilot study. *Crit Care Med.* 2010;38:419–427. [PubMed: 19915454]

43. Misak CJ. The critical care experience: a patient's view. *Am J Respir Crit Care Med.* 2004;170:357–359. [PubMed: 15105165]

44. Misak CJ. ICU psychosis and patient autonomy: some thoughts from the inside. *J Med Philos.* 2005;30(4):411–430. [PubMed: 16029990]

45. Hofhuis JG, Spronk PE, van Stel HF, et al. Experiences of critically ill patients in the ICU. *Intensive Crit Care Nurs.* 2008;24:300–313. [PubMed: 18472265]

46. Black P, Boore JR, Parahoo K. The effect of nurse-facilitated family participation in the psychological care of the critically ill patient. *J Adv Nurs.* 2011;67:1091–1101. [PubMed: 21214624]

47. Girard TD, Pandharipande PP, Carson SS, et al. Feasibility, efficacy, and safety of antipsychotics for intensive care unit delirium: the MIND randomized, placebo-controlled trial. *Crit Care Med.* 2010;38:428–437. [PubMed: 20095068]

48. Jones C, Griffiths RD, Slater T, et al. Significant cognitive dysfunction in non-delirious patients identified during and persisting following critical illness. *Intensive Care Med.* 2006;32:923–926. [PubMed: 16525845]

49. Hopkins RO, Key CW, Suchyta MR, et al. Risk factors for depression and anxiety in survivors of acute respiratory distress syndrome. *Gen Hosp Psychiatry.* 2010;32:147–155. [PubMed: 20302988]

50. Jubran A, Lawm G, Kelly J, et al. Depressive disorders during weaning from prolonged mechanical ventilation. *Intensive Care Med.* 2010; 36:828–835. [PubMed: 20232042]

51. Sukantarat KT, Williamson RC, Brett SJ. Psychological assessment of ICU survivors: a comparison between the Hospital Anxiety and Depression scale and the Depression, Anxiety and Stress scale. *Anaesthesia.* 2007;62:239–243. [PubMed: 17300300]

52. de Miranda S, Pochard F, Chaize M, et al. Postintensive care unit psychological burden in patients with chronic obstructive pulmonary disease and informal caregivers: a multicenter study. *Crit Care Med.* 2011;39:112–118.

53. Davydow DS, Gifford JM, Desai SV, et al. Depression in general intensive care unit survivors: a systematic review. *Intensive Care Med.* 2009;35:796–809. [PubMed: 19165464]

54. Dowdy DW, Bienvenu OJ, Dinglas VD, et al. Are intensive care factors associated with depressive symptoms 6 months after acute lung injury? *Crit Care Med.* 2009;37:1702–1707. [PubMed: 19357507]

55. Dantzer R, O'Connor JC, Freund GG, et al. From inflammation to sickness and depression: when the immune system subjugates the brain. *Nat Rev Neurosci.* 2008;9:46–56. [PubMed: 18073775]

56. Abrams TE, Vaughan-Sarrazin M, Rosenthal GE. Preexisting comorbid psychiatric conditions and mortality in nonsurgical intensive care patients. *Am J Crit Care.* 2010;19:241–249. [PubMed: 20436063]

57. Davydow DS, Russo JE, Ludman E, et al. The association of comorbid depression with intensive care unit admission in patients with diabetes: a prospective cohort study. *Psychosomatics.* 2011;52:117–126. [PubMed: 21397103]

58. Davydow DS, Hough CL, Russo JE, et al. The association between intensive care unit admission and subsequent depression in patients with diabetes. *Int J Geriatr Psychiatry.* 2011.

59. Desai SV, Law TJ, Needham DM. Long-term complications of critical care. *Crit Care Med.* 2011;39:371–379. [PubMed: 20959786]

60. Bienvenu OJ, Neufeld KJ. Post-traumatic stress disorder in medical settings: focus on the critically ill. *Curr Psychiatry Rep.* 2011;13:3–9. [PubMed: 20981585]

61. Jubran A, Lawm G, Duffner LA, et al. Post-traumatic stress disorder after weaning from prolonged mechanical ventilation. *Intensive Care Med.* 2010;36:2030–2037. [PubMed: 20661726]

62. Roberts BL, Rickard CM, Rajbhandari D, Reynolds P. Factual memories of ICU: recall at two years post-discharge and comparison with delirium status during ICU admission—a multicentre cohort study. *J Clin Nurs.* 2007;16:1669–1677. [PubMed: 17727586]

63. Treggiari MM, Romand JA, Yanez ND, et al. Randomized trial of light versus deep sedation on mental health after critical illness. *Crit Care Med.* 2009;37:2527–2534. [PubMed: 19602975]

64. Peris A, Bonizzoli M, Iozzelli D, et al. Early intra-intensive care unit psychological intervention promotes recovery from posttraumatic stress disorders, anxiety and depression symptoms in critically ill patients. *Crit Care.* 2011;15:R41.

65. Inouye SK, Bogardus ST, Charpentier PA, et al. A multicomponent intervention to prevent delirium in hospitalized older patients. *N Engl J Med.* 1999;340:669–676. [PubMed: 10053175]

66. Hunter BC, Oliva R, Sahler OJ, et al. Music therapy as an adjunctive treatment in the management of stress for patients being weaned from mechanical ventilation. *J Music Ther.* 2010;47:198–219. [PubMed: 21275332]

67. Jones C, Bäckman C, Capuzzo M, et al. Intensive care diaries reduce new onset posttraumatic stress disorder following critical illness: a randomised, controlled trial. *Crit Care.* 2010;14:R168.

68. Williams SL. How telling stories helps patients to recover psychologically after intensive care. *Nurs Times.* 2010;106:20–23. [PubMed: 20222486]

69. Frank AW. The standpoint of storyteller. *Qual Health Res.* 2000;10:354–365. [PubMed: 10947481]

70. Tanner DE. The narrative imperative: stories in medicine, illness and bioethics. *HEC Forum.* 1999;11:155–169. [PubMed: 11184852]

71. Räholm MB. Uncovering the ethics of suffering using a narrative approach. *Nurs Ethics.* 2008;15:62–72. [PubMed: 23268447]

72. Cuthbertson BH, Rattray J, Johnston M, et al. A pragmatic randomised, controlled trial of intensive care follow up programmes in improving longer-term outcomes from critical illness. The PRACTICAL study. *BMC Health Serv Res.* 2007;7:116. [PubMed: 17645791]

73. Schandl AR, Brattström OR, Svensson-Raskh A, et al. Screening and treatment of problems after intensive care: a descriptive study of multidisciplinary follow-up. *Intensive Crit Care Nurs.* 2011;27:94–101. [PubMed: 21334207]

74. Pattison NA, Dolan S, Townsend P, Townsend R. After critical care: a study to explore patients' experiences of a follow-up service. *J Clin Nurs.* 2007;16:2122–2131. [PubMed: 17313538]

75. Wang Y, Ma PL, Liu JT, et al. Relationship between severity of illness and mental status in ICU conscious patients: a nationwide multi-center clinical study [in Chinese]. *Zhonghua Yi Xue Za Zhi.* 2008;88:1450–1453. [PubMed: 18953848]

76. Abrams TE, Vaughan-Sarrazin M, Rosenthal GE. Preexisting comorbid psychiatric conditions and mortality in nonsurgical intensive care patients. *Am J Crit Care.* 2010;19(3):241–249. [PubMed: 20436063]